FitBuddy – AI Fitness Plan Generator Using Gemini Models

Project Specification & Design Document

1. Project Overview

FitBuddy is an AI-powered fitness application that generates personalized workout and fitness plans based on a user's age, gender, fitness level, goals, available equipment, preferred workout duration, and other inputs. The system uses Google Gemini models to analyze the user's requirements and generate customized workout recommendations, exercise descriptions, weekly schedules, and basic nutrition suggestions.

2. Objectives

• Generate personalized fitness plans using AI.
• Recommend exercises according to the user's fitness goal and experience.
• Create structured daily/weekly workout schedules.
• Provide exercise instructions and safety tips.
• Allow users to modify their fitness goals and regenerate plans.
• Provide an easy-to-use web/mobile interface.

3. Main Features

User Profile: Name, Age, Gender, Height, Weight, Fitness level, Fitness goal, Available equipment, Workout duration.
AI Fitness Plan Generator: Beginner/intermediate/advanced plans, Weight-loss, Muscle-building, Strength-training, General fitness, Home-workout plans.
Gemini AI Integration: Gemini receives structured user information and generates: Workout routines, Exercise names, Sets and repetitions, Rest periods, Weekly schedules, Exercise alternatives, General fitness guidance.
Progress Tracking: Completed workouts, Weight tracking, Workout history, Goal progress.
AI Chatbot: Users can ask questions such as: 'Give me a 30-minute home workout.', 'Replace squats with another exercise.', 'Create a beginner workout for Monday.', 'How should I structure my rest day?'

4. Suggested Technology Stack

Component	Technology
Frontend	React.js / HTML, CSS, JavaScript
Backend	Python Flask / FastAPI
AI	Google Gemini API
Database	Firebase / MongoDB / MySQL
Authentication	Firebase Authentication
Hosting	Firebase / Vercel / Render
API	REST API

5. Example Gemini Prompt

```
You are FitBuddy, an AI fitness planning assistant.

Create a 7-day fitness plan using the following user information:
Age: 22 | Fitness Level: Beginner | Goal: Weight Loss | Workout Duration: 30 minutes |
Equipment: No equipment | Location: Home

For each day provide: Exercise name, Sets, Repetitions or duration, Rest period, Short
instructions.
Include rest/recovery days and general safety guidance. Do not diagnose medical
conditions.
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6. Database Design

Users: user_id, name, email, age, gender, height, weight, fitness_level, fitness_goal, equipment
Workout Plans: plan_id, user_id, date, exercise_name, sets, repetitions, duration, rest_time, instructions
Progress: progress_id, user_id, date, weight, completed_workout, notes

7. Project Workflow

Register/Login → Create Fitness Profile → Select Fitness Goal → Enter Workout Preferences → Send Data to Backend → Gemini Generates Plan → Display Personalized Plan → User Completes Workout → Save Progress → Regenerate/Adjust Plan

8. Future Enhancements

AI-based food/meal suggestions • Exercise form analysis using computer vision • Integration with smartwatches • Voice-based fitness assistant • Automatic workout difficulty adjustment • Progress charts and analytics • Multilingual fitness assistant • Integration with Google Fit or other health platforms.

9. Project Abstract

FitBuddy – AI Fitness Plan Generator Using Gemini Models is an AI-based fitness assistance system designed to generate personalized workout plans according to individual user requirements. The application collects information such as age, fitness level, fitness goals, workout duration, and available equipment. This information is processed through a backend system and provided to a Gemini model, which generates a structured fitness plan containing exercises, sets, repetitions, rest periods, and general guidance. Users can track their completed workouts and interact with the AI assistant to modify or regenerate their plans.

This is for informational purposes only. For medical advice or diagnosis, consult a professional. AI responses may include mistakes.